

CSE 402 : Numerical Analysis, Simulation and Modeling Project

Title

Molecular Simulation on quantum computing systems : A generative approach



Nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical, and by golly it's a wonderful problem, because it doesn't look so easy.

— *Richard P. Feynman* —

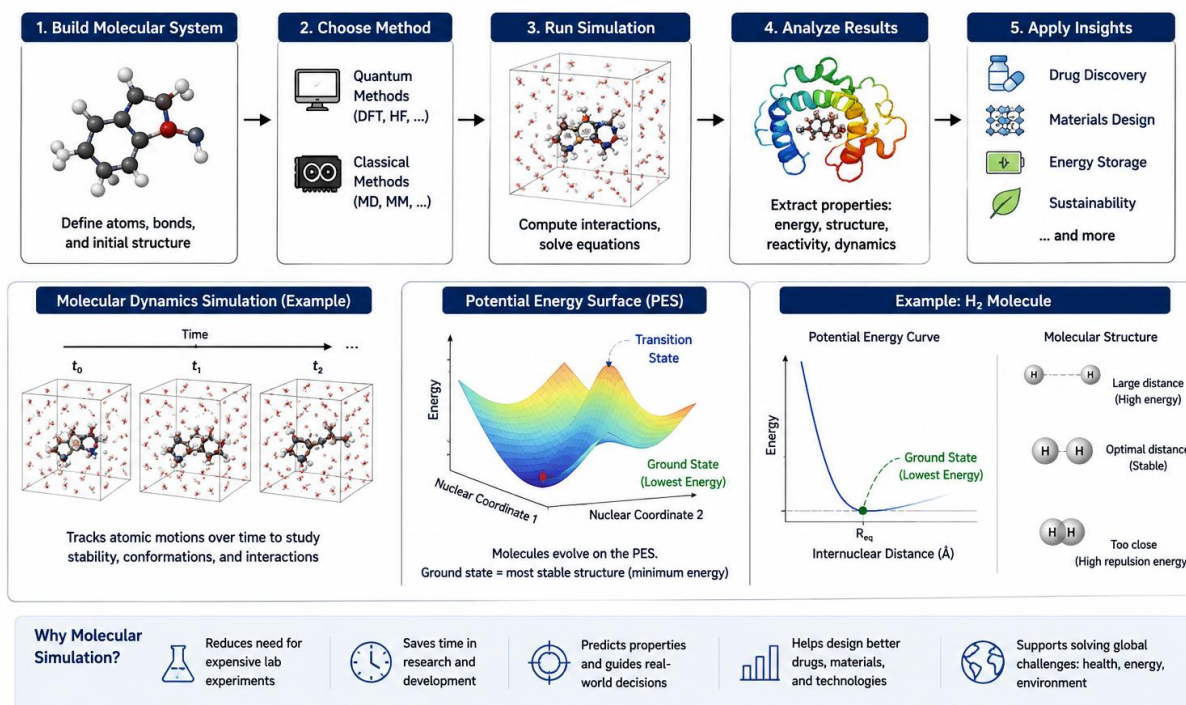
Background and Motivation

Molecular simulation is essential for understanding how molecules behave and interact, enabling advances in **drug discovery, materials science, energy storage, and chemical engineering**. Accurately predicting molecular properties can reduce the need for expensive and time-consuming laboratory experiments.

Among these properties, the **ground-state energy** is particularly important because it represents the most stable energy configuration of a molecular system and knowledge of the ground-state energy helps us understand molecular stability, chemical bonding, reaction behavior, and other important physical properties.

Molecular Simulation

Computational methods used to study and predict the behavior of molecules at the atomic and electronic levels



However, accurately computing the ground-state energy of complex molecules becomes computationally expensive for classical computers as the system size increases. Classical computers must store and process an **exponentially increasing amount of information** to accurately represent these states.

Even powerful supercomputers can struggle with the computational and memory requirements of realistic molecular simulations. Approximations can reduce the cost, but may sacrifice accuracy, especially for strongly correlated quantum systems.

This motivates the use of quantum computing techniques such as the **Variational Quantum Eigen solver (VQE)** to efficiently estimate molecular ground-state energies.

Literature Survey

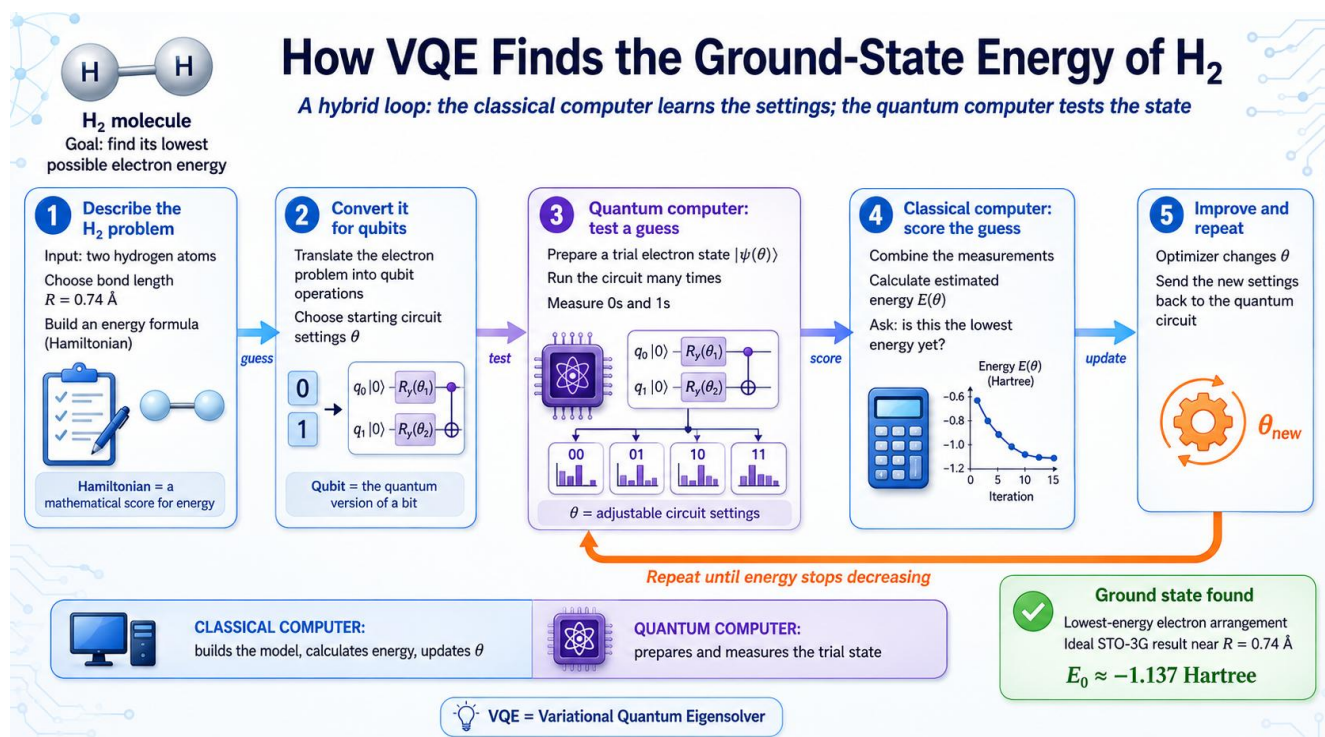
Variational Quantum Eigensolver (VQE)

[<https://www.nature.com/articles/ncomms5213>] (base paper)

[<https://www.sciencedirect.com/science/article/pii/S0370157322003118>] (survey paper)

is a **quantum simulation technique**, used to simulate complex chemical systems, which can mimic the natural superposition, entanglement and quantum tunneling using a quantum computer.

VQE Workflow



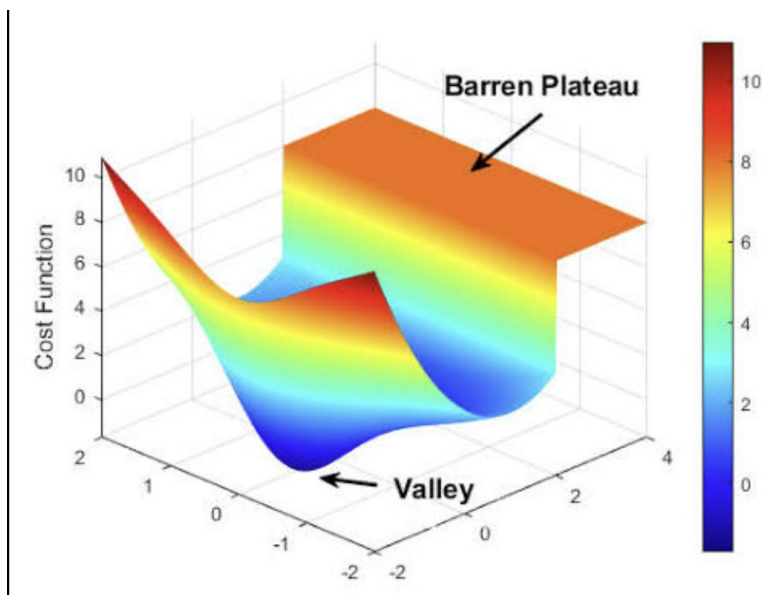
- 1) Encode the problem in the quantum system -> 2) Quantum Computer Guess ->
- 3) Classical Computer checks if the guess improves or converges [repeat until convergence]

Key VQE Challenges

- 1) Initial good guess (called Ansatz's in quantum system) , needs domain expertise to initialize quantum parameters
- 2) Optimization landscapes may become flat, harder to optimize (This event is called Barren Plateau Problem)

Optimization landscapes may become flat, harder to optimize, proved in a paper, named

Barren plateau [<https://arxiv.org/pdf/2405.00781>], causes the energy space to flat, gradient-variance becomes 0, difficult to optimize sometimes.

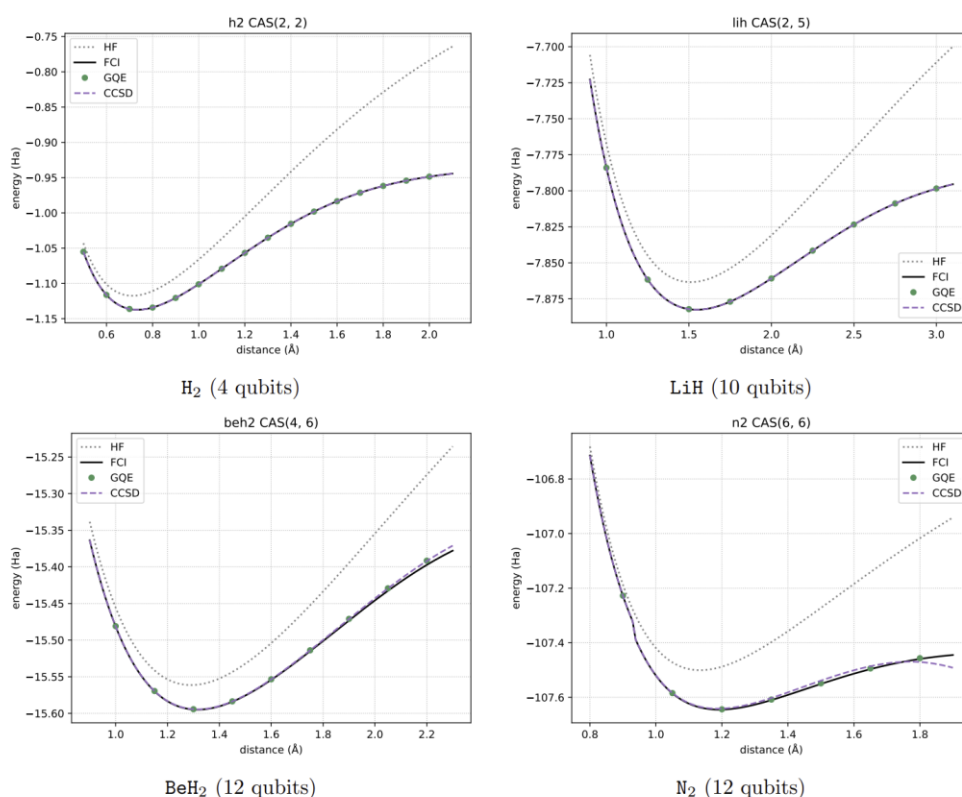


Suppose, this is the landscape and if you start the optimization step starting at a flat area, it won't move towards the lowest valley.

To mitigate the problem, several approaches have been tried.

One of them is [Generative Quantum Eigensolver \(arXiv:2401.09253\)](#) Which tries a **GPT-2 style architecture** and optimizes parameters outside the quantum computer which doesn't face this problem.

GQE demonstrates its feasibility to find ground state energy of H₂, Li-H, BeH₂ and N₂ molecules. If we look at the following graph, we see GQE converges more quickly and finds better solutions than classical well known methods used by chemical researchers. (HF, FCI, CCSD).



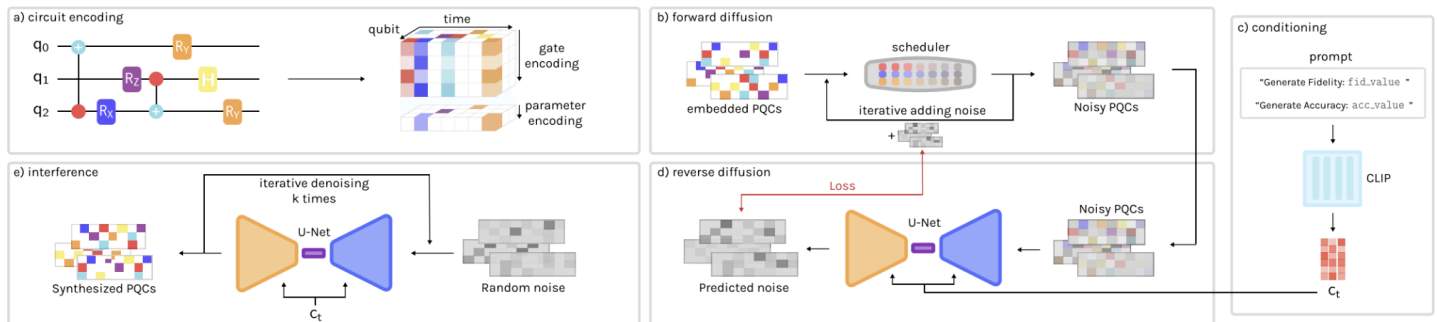
Open Research Gaps mentioned in GQE paper

- Scaling beyond small active-space molecular systems remains underexplored and should be evaluated with explicit resource limits.
- Cross-system transfer remains open: a model trained on one geometry or molecule may reuse useful circuit patterns on related targets

Diffusion-Based Ansatz Generation

(Solution to Problem 1)

To alleviate human labor, Diffusion Modeling has been tried for new Parameterized Quantum Circuit (Guess/Ansatz's) Generation that may supply a good guess initially : <https://arxiv.org/pdf/2505.20863>



Proposed Methodology

Combining both solution into one package

- 1) GPT-2 style ansatz generation is autoregressive, diffusion style conditional generation can be tried.
- 2) Exploring the scaling behavior
- 3) transfer learning.

Potential Publication Venue

Website: <https://aaai-qiml.github.io/> (a possible venue if the final study demonstrates sufficient novelty).

Second AAAI Symposium on Quantum Information & Machine Learning (QIML)

Bridging Quantum Computing and Artificial Intelligence

Areas of Interest

Quantum algorithms for classical AI tasks

Quantum-inspired machine learning techniques (QML)

Quantum Reinforcement Learning

Quantum Neural Networks & Graph Neural Networks

Quantum Generative Models

Variational Quantum Classifiers and Regressors

Quantum data encoding methods

Classical AI/ML for quantum systems

Scalability and hardware considerations

Applications of QC and QML

Hybrid Quantum-Classical (QC+HPC) Computing

Ethical considerations in QC and QML

- Paper Submission Deadline: 1st September, 2026
- Decision Notification : 15th, September, 2026
- Camera Ready Deadline: 29th, September, 2026